

AI Ethics: Autonomous Driving

By Christoph Lütge, Director, TUM Institute for Ethics in Artificial Intelligence (IEAI)

The development and implementation of autonomous vehicles is rising fast, while the scrutiny of some ethical issues fall behind. Therefore, this course module gives an overview of key ethical considerations relating to autonomous driving. In particular, the course will elaborate on the Trolley Problem, a thought experiment that has dominated the scientific debate in the field of autonomous driving ethics in the past. It will be highlighted why a move away from this concept and towards the ethics of risks is necessary and will introduce a first approach of doing so. Furthermore, the course will summarize some ethical recommendations that were developed by expert groups for industry and policymakers in the automotive sector. Lastly, the course will sketch the current status of legal frameworks.

Lecture Transcript

0:05 Hello, everyone. My name is Christoph Lütge. I'm a professor of Business Ethics at the Technical University of Munich. And I'm also the Director of the Institute for Ethics in Artificial Intelligence. I have been working on many aspects of AI ethics for years. But one aspect that we have special expertise in is the ethics of autonomous driving. I will come back to this in the course of my lecture. But I have been, for example, a member of the worldwide first ethics committee for autonomous driving in 2016, and 17, with the German Ministry of Transport. And this is one of the topics that my institute is also working on a lot, as you will also see, in the course of this lecture, I will give some examples. It is a topic that is both a technical topic and a non technical one. And I think this makes it very interesting. Same as with other AI ethics topics, but it seems that autonomous driving is something that is of great interest to many people, even if you only see a partial implementation already of it. But yeah, I think we should start at this point. So the development of autonomous vehicles is rising fast. We have a number of both technical and technological developments and progress that we see on many continents. So both in Europe, as well as in America and Asia. And we have also seen a number of firsts in incidents in which people were killed. So such as the Uber crash in Arizona, in 2018, where a woman was killed, who was walking across the street with her bicycle. And we have other incidents seen. So the question is, do ethical considerations fall behind? I wouldn't say broadly that they do. But it is something that needs to be paid attention to. What is in general at stake when we talk about ethics of autonomous driving? Here is in the chart, you see the total number of motor vehicle deaths in the US, starting from the beginning of the 20th century. And there has been a decline since the 1970s. But it is still a quite considerable number of around 35 or even 40,000 people that get killed each year. So this is something that is a problem in itself, and is something that needs to be paid attention to if there's the possibility to reduce these numbers, that would be an ethical improvement in itself and talking about this in German context. This would mean for Germany, around 3000 deaths that we could avoid or reduce at least as we have seen with introduction of other measures throughout the many years. So the point is that the introduction of autonomous vehicles, if they could significantly reduce, or I won't say totally avoid these deaths, that would be an

AI ETHICS

ethical advantage in itself. But of course, this is not the only consideration at stake. Autonomous vehicles can cause accidents.

4:33 I already made reference to the accident that happened in Arizona in 2018. Where a self driving vehicle hit a bicyclist, a woman, that was actually not riding on the bicycle, but we are pulling it over the street. And the question is from a broad ethical point of view, without going into all the details, is the technology mature enough and or are there ethical rules missing? For example, in this case, it was clear that the driver had not paid attention to the road. The vehicle had detected the woman with a bicycle in advance, but the braking system had not been activated for several reasons. And again, this is a longer story here, but there is a question - How should we set the rules? What rules should be given to the drivers of these vehicles? And so on. So, what can we do about this? Manufacturers often react to accidents that happen in a way like this, for example, Google's in 2016 said, "Well, we clearly bear some responsibility" in response to an accident, where a self driving car hit a bus, but it's not clear what that entails. In the Tesla crash in 2016, Tesla the company said, "the 'customer' is always in control and responsible" and that is, is certainly that that is important, but how we cannot always go on like this. If self driving cars are getting, getting more and more on the streets, getting deployed more and more, we will have to do something about the responsibility and the rules as well. And about the Uber crash that I mentioned, there was actually in the end, there was no basis found for criminal liability against the corporation. And again, these are at the moment, these are some cases. We also of course, see a crash and accidents with traditional cars, a lot of them as I mentioned. But if these cars get widely deployed, and as the technology is advancing more and more, we will have to deal with these, with these problems and with the possibility of these accidents, and we will have to do something about it. So, when it comes to autonomous driving, what are we actually talking about from a technical point of view? So here, you see the different levels of autonomous driving. These are from level zero to level five, there are slightly different classifications internationally, but this is probably the one that's used most and the differences are not that big, actually, between the different classification systems. So, the point is that the blue levels to the left, zero, one and two, are those levels where the driver is just supported by features of the car. So and which mean for example, like automatic lane keeping, or automatic braking but here, the driver must constantly supervise the support features, must steer, brake and accelerate as needed to maintain safety. So that's what the human being has to do in these levels. These levels are seen as rather unproblematic, highly automated driving, it's often called whereas the levels to the right levels three, four and five, those in which automated driving features will not require the human being to take over driving. In especially in the very high levels four and five, it means that over the entire course of the trip, the car can drive basically on its own, the it's not encoded driver, the human being is not required to supervise the system.

9:49 Level three is the crucial step ahead here actually, it will already mean that the car is taking over control. But if the car requests with some warning time ahead, if the car requests you to take, as a human being to take control again, then you are the one who must drive again. That's why this is there, some blue and some green in level three. And that's where we currently stand. So, so far, almost all of the cars that have automated features are on level two, there are some that have for specific use cases a level three classification, which is debatable and it's being discussed, but the big step will be to take this further beyond three and to levels four and maybe five. So, again, from an ethical point of view, in a nutshell, that means there are advantages from an ethical point of view of autonomous driving: avoiding accidents, reducing physical and mental stress, saving time, more

AI ETHICS

convenience, of course. And again, this cannot be stressed too much, since according to all the statistics, more than 90% of all accidents happen due to human errors. And so if we can reduce this significantly, this would mean saving a lot of lives, but we will need some rules for these cars. So if we need ethical rules, then what can ethics say about the autonomous driving and one key concept that must be mentioned is the trolley problem. But I will also, I will explain this in a short moment. But I will also then go beyond the trolley problem because I believe it's it's, it's also too much. There was too much discussion of the trolley problem in recent time.

12:24 So what is the trolley problem? The trolley problem was originally only a thought experiment by philosophers, which it was devised in the 60s. And it went like this: a tram is out of control and threatens to overrun five people. By moving a switch, the tram can be diverted to another track. Unfortunately, there's another person on the track. And that can the death of one person be accepted by moving the switch in order to save the lives of five people. And this is maybe the best known theoretical case of a moral dilemma. It's often pictured like this. And it originally was a thought experiment to sharpen our intuitions about what kind of ethical position would we adapt? Would we be a utilitarian? Utilitarians would say you should maximize happiness of all, so that would mean saving more people is always preferred. Not not sacrifice. So to sacrifice one person would be better than sacrificing five. Where versus the other point of view would be an individual case decision. If you are and if you're not a utilitarian, you might refuse to move the switch to reroute the train. This is a value judgment that has been taken. I don't want to decide consciously against one person and in the course kill this person.

14:16 So while this was originally only a thought experiment with actually no connection to autonomous driving whatsoever, I mean, there was no there was no research on that done in the 60s of course. The moral machine experiment then became very popular. As in then the Moral Machine experiment took the Trolley Dilemma one step further if you want and the moral machine is an online platform developed from a research project at the MIT and this project, I mean, this was opened around 2000, I don't have the exact year but 15 or 16, I think, which, which the aim was to identify broad view on how machines should solve moral dilemmas and offering a forum where the public can discuss ethically problematic scenarios. And there was an explicit view now on autonomous driving. So there's the, there is the link, which you can take, and you go there, and you do the experiment yourself. So these always look like this. And thank you for the people who I had the pleasure of talking to, the people behind the Moral Machine, it was a great experience, like Iyad Rahwan, and, and others, The Moral Machine experiment here, this is this is one scenario, and you can come up with a big number of scenarios here. Here, it says, sort of there's a car without passengers and autonomous cars coming along the street and should it kill the five people in the as in the left picture? Or should it be diverted to the right and kill only this one person? And you can judge what would be right from your point of view. There's a lot of variations here. For example: here five persons or one dog; there's also something with cats here, for example; you have dogs and cats versus three people; three younger persons you see on the left, and three older people on the right; three women versus three men; passengers in the car now so far, we've only had an empty car here, passengers in the car and versus pedestrians; should the car go against the wall or should it be diverted and killed the pedestrians? This is one case actually, which I find particularly problematic since the people inside the car are usually much better protected than people outside. So here the assumption is they are killed with 100% certainty and which is often now been almost all cases not not true. Again, pedestrians also there, there you see already, it doesn't make a difference whether you interfere with the the course that the car was taking whether it was going straight on, should you interfere and do something or should should rather let it go. I won't decide on this. This is for you to decide if you

AI ETHICS

take the Moral Machine experiment online. This for example is a very interesting case where you see some people crossing the street on, on where it's not allowed. So if their traffic lights are red, and so they are prohibited from crossing the road at that point, does that make a difference? And as we will see, actually this makes a great difference for many people. They would say well if someone is not following the rules that that is an important thing to take into account. And because I mean the car has to follow some rules and if you continuously reckon with people who aren't who are not doing so, then you will have a problem simply with the program. But it's this is this is one of the one of the problems that has been fined and if you can, how realistic this is if you can find out those opposition's those are bank measures I will leave that open or if you if you really can identify a criminal that is also of course, an idealization. But we are always talking about idealizations in this context.

19:44 So some of the key findings to summarize these of the Moral Machine experiment were, again, I won't go into all the details here. This was in an article in Nature by Awad and colleagues. That, in particular, there is a preference for action, whereas some of let's say, let's say this way, the key findings, let's start with the right side. So, with girl and boy, there was no not much difference as you see here. Male doctor, female doctor, there was not much difference. There was a kind of when you compare young people and old people, girl and boy, there is at least in this context, in this cultural context, and also has been done in different contexts, there was a preference for for sparing young people characters against older ones. And there was a there was a clear preference for sparing people over animals whereas, the sex of the people did not make much difference actually. So, again, then there is if you look to the left, there is a preference for, for law abiding action. There is a preference for rather, sparing more people and, and, as one would expect, in a utilitarian framework, if you can save more people, that's an action that you will take. Again, human beings, also human beings were approved preferred over pets. Whereas, pedestrians and passengers, there's a slight preference here, but not the big one. Higher status is preferred, to some extent here, of course question is how you can identify those people. Whereas action and inaction does not make much difference. So, these, these were the findings, again, again, summarized here. There are some complications, if you introduce different cultural backgrounds and the the Moral Machine problem has been, has been performed. I mean, the the data that have been used, have been traced back to different countries to different regions of the world, different cultural backgrounds. And there are some differences were found, especially that, that people from so-called Western countries prefer children rather than older people. And it looks like that it's different in Eastern countries. That's at least what what the data says. And this can be made even more complex. I won't go into into into the details here. Again, sparing the lawful, for example, is something that's very much preferred in in the Eastern context. And sparing pedestrians is also is also much more preferred here. Again, this can be traced back to more detail, but this is for you to find out. One final point with regard to the Trolley Dilemma, the legal situation in Germany at least, is has a particular point here, because some years ago, actually, already 15 years ago, there was there was a Supreme Court constitutional court decision about whether it is legal to shoot down a hijacked passenger plane. And there was a law that had been passed, but it was judged unconstitutional because the Constitutional Court said this is not in accordance with human dignity. So what does that mean for the the trolley problem? This is something we had to deal with when it came to the first ethics committee on autonomous driving, which I will turn to in a moment.

25:01 So first, let's now talk. Let's now, go beyond the trolley problem and discuss the ethics of risk, because the question is, what or are there any shortcomings of the trolley problem? While it certainly has a number of benefits, it raises the reflection and understanding of normative theories that's what it was already originally

AI ETHICS

devised for. There are several severe shortcomings, especially in the context of autonomous driving. So it's possible its consequences as a cert. This is something that you can change in, in the in the settings of the Moral Machine, but it's not it's not in the in the standard case, at least. There are a limited number of options available and there's also a lot of contextual information missing. I mean, this is this is always the case with with these kinds of models, models are abstracting from contextual information, otherwise, they wouldn't be models. But still, the question is, then how useful can they be? How relevant for practice can they be? And that's why a number of scholars have in the past years, called for reframing the debate, like disarming the trolley problem, away from trolley problems, trolled by the trolley problem and so on. So, so, we see that there is that there is a clear tendency to go beyond that, and not just from a scholarly perspective, but also from a policymaker perspective, because, as for example, the European Commission Expert Group in 2020 has said, we need to talk about dilemmas in crash avoidance and not just not just about the dilemmas in an unavoidable crash situation situation. These situations cannot be totally avoided, but more the vast majority of cases can be dealt with by a lot of a number of measures, technological, non technological ones, in order to avoid crashes as much as possible. That means reframing the debate from a trolley perspective towards a risk based perspective, and risks are something that is there that is that we are always faced with in our ordinary lives. In our everyday lives, we have to deal with risks, we are taking risks every day, when we go out on the street or drive a car or ride a bicycle. And that is that is completely different from these very rare cases of trolleys. And for example, what kind of risks are we then faced with like for example, like this, how close should an autonomous car get towards here in this case, a truck and a bicycle. And on the left side, you see a kind of what you could call an ego vehicle that here the risk would be more on the on the vehicle and to the right and there would be more risk on the cyclist and because in the right picture, the car would be programmed to go closer to the cyclist, whereas in the left, it would rather have to go closer to the truck. So there we do not make a decision about lives, but we are putting some people slightly more at risk than others.

29:20 And that's also of course, important decisions that have to be taken when it comes to programming autonomous cars. From an ethical point of view, we need to ask then how can we achieve a fair distribution of risks? For example, in the trajectory planning of automated or highly automated vehicles. So they have to plan their trajectories in the programming: Should it go this way? Should it go that way? How close should it come to different cars? And there are a lot of questions which are technical ones but also involve ethical questions. You need to calculate the risk of the autonomous vehicles. You need to distribute somehow the risks and you need to discuss what counts as a fair distribution of such risks. And that might even change over the course of time. When we have for example, new technological developments, but maybe also other discussions in ethics, what we accept as, as fair or not fair in society. There is a project here at the Institute, I invite you to have a look at the at our institute webpage, it's called the ANDRE Project that discusses these questions in more detail. So, now, we have talked a bit about the background of ethical questions in connection with autonomous driving. What have been developments in respect with respect to guidelines, because guidelines and and regulation now, recently have been have been developed in the past few years. So, in 2017, the German Ethics Code for Automated and Connected Driving was presented was the German Federal Ministry of Transport and Digital Infrastructure. In that time, it included 20 ethics guidelines. For example, one that said personal injuries always take precedence of a damage to property. So, which means that yeah, damage to property must rather be taken, much rather be accepted, then then personal injuries. Another one said that liability passes from when it comes to autonomous driving, if the car was going on its own or the driver was not in control and the liability passes from the user

AI ETHICS

owner to the manufacturer and the operator of the software. And also, one guideline that said general programming to reduce the number of personal injuries should be permissible. But in some more detail, so this was the chairperson Udo Di Fabio was a former judge of the Constitutional Court of Germany. And there were other experts from industry, academia and civil society. I will not go into all the details here. But the guidelines clearly said that there is an ethical benefit to autonomous driving. We should implement it because it can help us achieve goals like reducing accidents, ethical goals. We should achieve a positive risk balance, again this risk perspective is also already there. We need to accept animal, property damage and what was particularly important here is the what is here number eight. In accident situations, which are unavoidable, there must be no qualification according to personal characteristics, such as age or sex or physical constitution or whatever. That would be prohibited. However, a general programming that reduces the overall number of personal injuries that can be justifiable, which is an important point of view, which means that you can program an overall reduction of injuries, but not base calculations on individual characteristics.

34:18 There are a number of other learnings here again, I won't go into all the details, but we are saying we have been saying more about the handover practices which must be clear and over from control. So handing over control from machine to person, the safe conditions of a car, how should they look like? Can they be standardized, and also what the what the role of education, what the changes in education are that are to be expected. This was taken further a few years later, by AI4People Automotive Committee, which in 2020, developed ethical guidelines specifically for the automotive sector. And it included in particular more concrete guidelines for companies or other organizations working in this sector. It was involved with human agency and oversight, technical robustness and safety, diversity, environmental well being as well as also accountability. And there were several recommendations and just to give you some ideas, we, there were recommendations on the policy level, as well as on the industry level. And one one was that, for example, there should be conditional override option that allows the control to be handed back to the driver. And this is this is controversial. But at the moment, we feel that it's not, not yet the time to do away with this override option. It might be seen differently in a number of years time. And, yeah, this override function should depend on the level of automation of the AVs. Again, there is also the recommendation to the right that responsible balancing of risks and estimated harm to reduce the number of personal injuries should be permitted for autonomous vehicles without discriminating against personal characteristics. That's what the German ethics code already said. What we could include is the type or speed of particular traffic participants and the impact angle under which a collision would occur. So, just just as some examples, what we can include, but others we could not. And there are more to find. There's the source to the right. The idea is then how can the recommendations be turned into into practice such as for example, compliance with regulations standards for cybersecurity management systems, and there have been standards developed already. There will have to be data protocol protocols implemented about who can can access which data and under which conditions and yeah, there are others here as well. So, some key takeaways, again, would be not just the balancing of risks is recommended. We should also, I think this is a very important general point, we should rather see the implementation of AVs in a more step by step manner rather than taking the full staircase by one leap, but rather see it as a step by step approach. And for that, we also need, by the way this was published in early 21, to develop a clear regulatory framework. And that was something that was I mean, companies have been developing some principles on their own, but we will need to discuss the legal frameworks for autonomous driving here in more detail and that there is this is a development that has been taken further just more recently,

AI ETHICS

especially in 21. Because there are ethical aspects, such as this ethical code on autonomous driving says, but there are also legal aspects.

38:56 And there we have, we have already some frameworks like the General Data Protection Regulation, we have consumer protection laws. But we also have for example, the upcoming EU AI act. So which has not yet been passed, but which is planned. And the general point of all this is to establish trust in the technology so that it will actually be used. In Germany, there was an act on autonomous driving that was passed in 2021 and which was, there was a line of continuation from the Ethics Committee to this to this act. It is called an interim solution until internationally harmonized provisions enter into force. And, again, without going into all the details this act is, we would like to make mobility of the future more diverse, safer, more environmentally friendly and user focused. It specifies an operating range for autonomous driving. So if a vehicle is an autonomous vehicle or is an autonomous driving function, it can perform a driving task independently within a specific specified operating range and has certain technical equipment. Yeah, there's more more details that you can have a look at this yourself and entails compliant costs for manufacturers for around 300,000 euros per year, and and others. In particular, what's interesting to see is the deployment scenarios that are specified here. Because the regulation here concentrates primarily on the public sector rather than the private one. The scenarios here proposed by the Ministry of shuttle services, people movers which are buses on established routes, which cannot go to the side of the road, hub2hub services, demand driven offers and off peak hours and, and others. So automated valet parking also. So these are the scenarios that are contemplated here. And again, there are more details here that I recommend you to have a look at. It will certainly be not the last regulation in this in this context. There are some duties, for example, that is that are asked from the owners, ensuring regular maintenance of the systems. This is of course, something that we have had before the Autonomous Driving Act. There is a lot about technical oversight, and what producers have to have to do, detecting manipulations of the vehicles and so on, and delivering data on requests to the Kraftfahrt-Bundesamt, which would be the agency that would have to look at and to guarantee this the safety of these technical systems. So again, this will certainly not be the last autonomous driving act, it is as it was called an interim act and we will certainly see more of these. In particular, what is very desirable, is a harmonization on the European or EU level.

43:07 So to conclude, we have seen one important step is to shift the scientific debate away from the trolley problem towards an ethics of risks. This is something that is currently being done, has been done so to some extent already in the last one or two years. The next step is to provide practical guidelines. So not just these very highly abstract ones, which have been important as well. But now we need more practical guidelines for the industry, for other organizations also, certainly for the state. And these must be from a legal and an ethical perspective. And finally, what is maybe the most important thing is to create interdisciplinary teams that are working on the development of these technologies, because many of the of the questions that turn up in the development of these autonomous vehicles are not just technical ones, but they are ethical ones or have an ethical aspect in there, which you only find out when you actually get to those questions in detail. Those are questions that you can solve them in advance. Like for example, this question how close should the car get to other to other vehicles. So that's what we need to what for example, companies working on these systems need to create teams with additional skills with skills that are not just technical ones, but non technical ones, who can then discuss with the engineers and programmers what the best, the the fair, solution would be for such technology. So that's, that's

AI ETHICS

all for today and I encourage you to have a look to have a further look at the literature that was mentioned. And thank you very much for today.